Elektrodynamika

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Elektrodynamika

poloha $t, \vec{r}$ $\quad x^\mu = \begin{bmatrix} ct \\ \vec{r} \end{bmatrix}$

zdroje $\rho, \vec{j}$ $\quad j^\mu = \begin{bmatrix} c\rho \\ \vec{j} \end{bmatrix}$

potenciály $\phi, \vec{A}$ $\quad A^\mu = \begin{bmatrix} \frac{1}{c}\phi \\ \vec{A} \end{bmatrix}$

EM pole $\vec{E}, \vec{B}$ $\quad F_{\mu\nu} = \begin{bmatrix} 0 & -\frac{1}{c}\vec{E} \\ \frac{1}{c}\vec{E} & \overleftrightarrow{B} \end{bmatrix} =$

$$F_{ij} = \varepsilon_{ijk} B^k \qquad = \begin{bmatrix} 0 & -\frac{1}{c}E_x & -\frac{1}{c}E_y & -\frac{1}{c}E_z \\ \frac{1}{c}E_x & 0 & B_z & -B_y \\ \frac{1}{c}E_y & -B_z & 0 & B_x \\ \frac{1}{c}E_z & B_y & -B_x & 0 \end{bmatrix}$$

hustota L. síly $\vec{f}$ $\quad \Phi = \begin{bmatrix} \frac{w}{c} \\ \vec{f} \end{bmatrix}$ ← hustota energie

invarianty $\quad \mathcal{L} = \frac{\varepsilon_0}{2} E^2 - \frac{\varepsilon_0 c^2}{2} B^2 = -\frac{\varepsilon_0 c^2}{4} F_{\mu\nu} F^{\mu\nu}$

$$\tilde{\mathcal{L}} = \varepsilon_0 c\, \vec{E}\cdot\vec{B} = \frac{\varepsilon_0 c^2}{8} F^{\kappa\lambda} F^{\mu\nu} \varepsilon_{\kappa\lambda\mu\nu}$$

Maxwellky

$\vec{\nabla}\cdot\vec{E} = \frac{1}{\varepsilon_0}\rho \qquad \vec{\nabla}\times\vec{B} - \frac{1}{c^2}\frac{\partial\vec{E}}{\partial t} = \frac{1}{\varepsilon_0 c^2}\vec{j} \qquad \nabla_\mu F^{\alpha\mu} = \frac{1}{\varepsilon_0 c^2} j^\alpha$

$\vec{\nabla}\cdot\vec{B} = 0 \qquad \vec{\nabla}\times\vec{E} + \frac{\partial\vec{B}}{\partial t} = 0 \qquad \nabla_{[\kappa} F_{\mu\nu]} = 0$

↑ antisymetrizace a cyklická permutace jsou až na prefaktor shodné kvůli antisymetrii $F_{\mu\nu}$

Potenciály

$\vec{E} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t}$

$\vec{B} = \vec{\nabla}\times\vec{A} \qquad F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$

Hustota Lorentzovy síly

$\vec{f} = \rho\vec{E} + \vec{j}\times\vec{B}$

$w = \vec{j}\cdot\vec{E} \qquad \Phi^\alpha = F^\alpha{}_\mu j^\mu$

Rovnice pro potenciály

$$\frac{1}{\varepsilon_0}\rho = \vec{\nabla}\cdot\vec{E} = -\vec{\nabla}\vec{\nabla}\phi - \vec{\nabla}\cdot\frac{\partial\vec{A}}{\partial t}$$

$$= -\Big[\underbrace{-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \Delta}_{\square}\Big]\phi - \frac{\partial}{\partial t}\left(\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \vec{\nabla}\cdot\vec{A}\right)$$

$$\Rightarrow \quad \square\phi = -\frac{1}{\varepsilon_0}\rho - \frac{\partial}{\partial t}\left(\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \vec{\nabla}\cdot\vec{A}\right)$$

$$\frac{1}{\varepsilon_0 c^2}\vec{j} = \vec{\nabla}\times(\vec{\nabla}\times\vec{A}) + \frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2} + \frac{1}{c^2}\frac{\partial}{\partial t}\vec{\nabla}\phi$$

$$= -\left[-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \Delta\right]\vec{A} + \vec{\nabla}\left(\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \vec{\nabla}\cdot\vec{A}\right)$$

$$\Rightarrow \quad \square\vec{A} = -\frac{1}{\varepsilon_0 c^2}\vec{j} + \vec{\nabla}\left(\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \vec{\nabla}\cdot\vec{A}\right)$$

$$\square = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \Delta$$

$$\Leftrightarrow \quad \frac{1}{\varepsilon_0 c^2} j^\mu = -\nabla_\nu\nabla^\nu A^\mu + \nabla^\mu(\nabla_\nu A^\nu)$$

Kalibrační volnost

$\vec{A} \to \vec{A}' = \vec{A} + \vec{\nabla}\chi$

$\phi \to \phi' = \phi - \frac{\partial\chi}{\partial t}$

$$\Rightarrow \quad \vec{B}' = \vec{\nabla}\times\vec{A}' = \vec{\nabla}\times\vec{A} + \overbrace{\vec{\nabla}\times\vec{\nabla}\chi}^{\overset{0}{\equiv}} = \vec{B}$$

$$\Rightarrow \quad \vec{E}' = -\vec{\nabla}\phi' - \frac{\partial\vec{A}'}{\partial t} = \vec{\nabla}\phi + \underbrace{\vec{\nabla}\frac{\partial\chi}{\partial t} - \frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\frac{\partial\chi}{\partial t}}_{=0} = \vec{E}'$$

$$\Rightarrow \quad A_\mu \to A'_\mu = A_\mu + \nabla_\mu\chi$$

Lorenzova kalibrační podmínka

$\frac{1}{c^2}\frac{\partial\phi}{\partial t} + \vec{\nabla}\cdot\vec{A} = 0 \qquad \nabla_\mu A^\mu = 0$

$\Rightarrow \quad \square\phi = -\frac{1}{\varepsilon_0}\rho \qquad \square\vec{A} = -\frac{1}{\varepsilon_0 c^2}\vec{j}$

+ máme dodatečnou volnost s podmínkou

$\square\chi = 0$

$\Rightarrow$ kompatibilní s Lorenzovou podmínkou

Coulombova kalibrace

$\vec{\nabla}\cdot\vec{A} = 0$

řešíme stejně jako v elektrostatice

$\Rightarrow \quad \Delta\phi = -\frac{1}{\varepsilon_0}\rho \qquad \square\vec{A} = -\frac{1}{\varepsilon_0 c^2}\vec{j} + \frac{1}{c^2}\vec{\nabla}\frac{\partial\phi}{\partial t}$

⇓

$$\phi = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(x')}{r(x|x')}dV$$

v jednom čase
v celém prostoročase $\Rightarrow$ nekauzální šíření, ale nevadí to, protože ϕ a $\vec{A}$ jsou neměřitelné
-> $\vec{E}$ a $\vec{B}$ už se šíří kauzálně

-> speciální případ, pokud $\rho = 0$
$\Rightarrow \quad \phi = 0$ } bezzdrojový problém

Weylova kalibrační podmínka

$\phi = 0 \xrightarrow[\text{zobecnění}]{} \bar{\phi} =$ cokoliv $\quad$ z $\underbrace{\phi \to \bar{\phi} - \frac{\partial\chi}{\partial t}}_{\text{není těžké vyřešit}}$

Zákony zachování (obecně)

$$\int_V \underbrace{w}_{\text{nějaká veličina}}\Big|_{t_k} dV - \int_V w\Big|_{t_z} dV + \int_{t_z}^{t_k}\int_{\partial V}\underbrace{\vec{w}}_{\text{výtok veličiny}}\cdot d\vec{S}\,dt = \int_{t_z}^{t_k}\int_V \underbrace{z}_{\text{zdroj veličiny = to, co vzniká; tj. hustota tvorby}} dV dt$$

$$\int_{t_z}^{t_k}\int_V \frac{\partial w}{\partial t}dVdt + \int_{t_z}^{t_k}\int_V \vec{\nabla}\cdot\vec{w}\,dVdt = \int_{t_z}^{t_k}\int_V z\,dVdt$$

$\Rightarrow \quad \frac{\partial w}{\partial t} + \vec{\nabla}\cdot\vec{w} = z$ ← zobecnění - dodává zdroj $\quad$ Rovnice kontinuity

$\Leftrightarrow \quad \nabla_\mu w^\mu = z \qquad w^\mu = \begin{bmatrix} cw \\ \vec{w} \end{bmatrix}$

Konvektivní proud: $\quad \vec{w} = w\,\vec{v}$ ← rychlost

Zákon zachování náboje

$\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot\vec{j} = 0 \qquad \nabla_\mu j^\mu = 0$

$$\frac{1}{\varepsilon_0}\left(\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot\vec{j}\right) = \frac{\partial}{\partial t}\cancel{\vec{\nabla}\cdot\vec{E}} + c^2\underbrace{\vec{\nabla}\cdot\vec{\nabla}\times\vec{B}}_{\equiv 0} - \cancel{\vec{\nabla}\cdot\frac{\partial\vec{E}}{\partial t}} = 0$$

=> důsledek Maxwellových rovnic

Bilance energie

Energie dodaná do EM pole:

$= -w = -\vec{j}\cdot\vec{E}$

↖ výkon

$\rho = \sum_k \rho_k \qquad \vec{j} = \sum_k \vec{v}_k\rho_k$

$\vec{f}_k\cdot d\vec{s}_k = \vec{f}_k\cdot\vec{v}_k\,dt = \rho_k(\vec{E} + \underbrace{\vec{v}_k\times\vec{B}}_{\text{je nulový}})\cdot\vec{v}_k\,dt$

$= \rho_k\vec{E}\cdot\vec{v}_k dt = \vec{E}\cdot\vec{j}_k\,dt$

$$\Rightarrow \quad -w = -\vec{j}\cdot\vec{E} = \varepsilon_0 c^2\left(-\vec{\nabla}\times\vec{B} + \frac{1}{c^2}\frac{\partial}{\partial t}\vec{E}\right)\cdot\vec{E}$$

$$= \varepsilon_0 c^2\left(\vec{\nabla}\cdot(\vec{E}\times\vec{B}) - \vec{B}\cdot(\vec{\nabla}\times\vec{E}) + \frac{1}{2c^2}\frac{\partial}{\partial t}\vec{E}^2\right)$$

$$= \frac{\partial}{\partial t}\left(\frac{\varepsilon_0}{2}E^2 + \frac{\varepsilon_0 c^2}{2}B^2\right) + \varepsilon_0 c^2\,\vec{\nabla}\cdot(\vec{E}\times\vec{B})$$

$$= \frac{\partial u}{\partial t} + \vec{\nabla}\cdot\vec{S}$$

hustota energie (u); Poyntingův vektor ($\vec{S}$) => hustota toku energie

$$\Rightarrow \quad -w = \frac{\partial u}{\partial t} + \vec{\nabla}\cdot\vec{S}$$

$\int_S \vec{S}\cdot d\vec{S}$ → výtok přes plochu

Bilance hybnosti

-hybnost dodaná do EM pole $\qquad$ časová změna hybnosti je síla

$= -\vec{f} = -(\rho\vec{E} + \vec{j}\times\vec{B})$

$$= -\varepsilon_0(\vec{\nabla}\cdot\vec{E})\vec{E} - \varepsilon_0 c^2(\vec{\nabla}\times\vec{B})\times\vec{B} + \varepsilon_0\frac{\partial\vec{E}}{\partial t}\times\vec{B}$$

$$= -\varepsilon_0(\vec{\nabla}\cdot\vec{E})\vec{E} - \varepsilon_0 c^2\underbrace{(\vec{\nabla}\cdot\vec{B})}_{\equiv 0}\vec{B} + \varepsilon_0\frac{\partial}{\partial t}(\vec{E}\times\vec{B})$$

$$+ \varepsilon_0\vec{E}\times(\vec{\nabla}\times\vec{E}) + \varepsilon_0 c^2\vec{B}\times(\vec{\nabla}\times\vec{B})$$

$$= \frac{\partial}{\partial t}(\varepsilon_0\vec{E}\times\vec{B}) - \varepsilon_0(\vec{\nabla}\cdot\vec{E})\vec{E} - \varepsilon_0\vec{E}\cdot\vec{\nabla}\vec{E} + \varepsilon_0(\vec{\nabla}\vec{E})\cdot\vec{E}$$

$$- \varepsilon_0 c^2(\vec{\nabla}\cdot\vec{B})\vec{B} - \varepsilon_0 c^2\vec{B}\cdot\vec{\nabla}\vec{B} + \varepsilon_0 c^2(\vec{\nabla}\vec{B})\cdot\vec{B}$$

$$= \frac{\partial}{\partial t}(\varepsilon_0\vec{E}\times\vec{B}) - \varepsilon_0\vec{\nabla}\cdot\left[\vec{E}\vec{E} + c^2\vec{B}\vec{B} - \frac{1}{2}(E^2 + c^2B^2)\overset{\text{pozn. si metriku}}{g}\right]$$

$$= \frac{\partial\vec{g}}{\partial t} + \vec{\nabla}\cdot\overleftrightarrow{T}$$

hustota hybnosti ($\vec{g}$); $\overleftrightarrow{T}$ ← Maxwellův tenzor napětí => hustota toku hybnosti

$T_{ij} = -\varepsilon_0\left[E_iE_j + c^2B_iB_j - \frac{1}{2}(E^2 + c^2B^2)g_{ij}\right]$

$$T^{\mu\nu}_{EM} = \begin{bmatrix} u & c\vec{g} \\ \frac{1}{c}\vec{S} & \overleftrightarrow{T} \end{bmatrix} \qquad c\vec{g} = \frac{1}{c}\vec{S}$$

$$T^{\mu\nu}_{EM} = \varepsilon_0 c^2\left(F^{\mu\kappa}F^{\nu\lambda}\eta_{\kappa\lambda} - \frac{1}{4}F_{\kappa\lambda}F^{\kappa\lambda}\eta^{\mu\nu}\right)$$

$$\nabla_\mu T^{\mu\nu}_{EM} = -\Phi^\nu$$

obecně lze formulovat

Obecně: $\quad \nabla_\mu T^{\mu\nu}_{celk.} = 0$ ← vyplývá z invariance vůči difeomorfismům

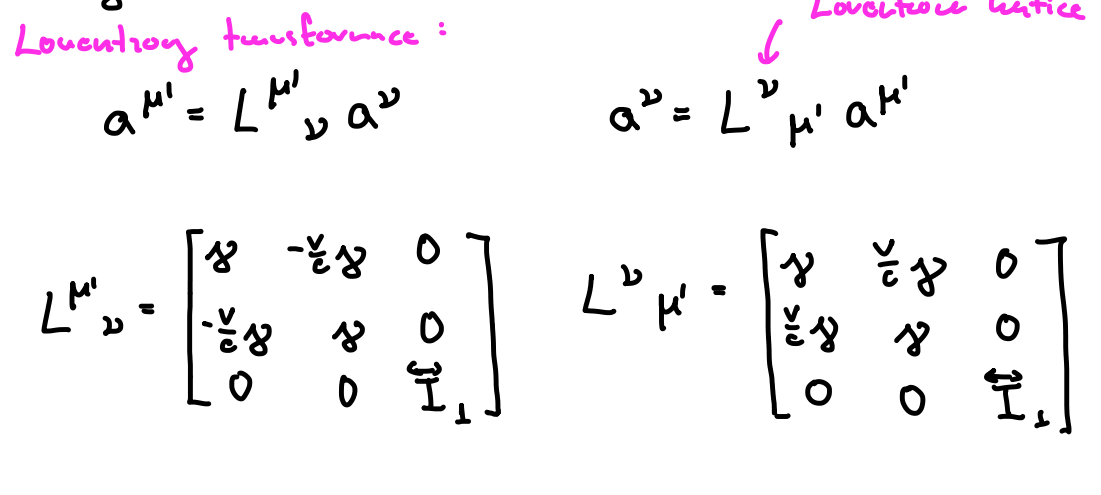
$$T^{\mu\nu}_{celk} = \underbrace{T^{\mu\nu}_{EM} + T^{\mu\nu}_{prach} + T^{\mu\nu}_{SP} + \dots}_{\text{všechny interagující členy}}$$

($T^{\mu\nu}_{SP}$ ← skalární pole)

$$\Phi^\nu = -\nabla_\mu T^{\mu\nu}_{EM} = \nabla_\mu T^{\mu\nu}_{ostatní}$$

Lorentzova transformace

- transformace mezi dvěma inerciálními soustavami



$\gamma = \cosh\vartheta$ ← rapidita $\qquad \frac{v}{c} = \tanh\vartheta$

Lorentzova transformace:

$a^{\mu'} = L^{\mu'}{}_\nu a^\nu \qquad a^\nu = L^\nu{}_{\mu'} a^{\mu'}$ ← Lorentzova matice

$$L^{\mu'}{}_\nu = \begin{bmatrix} \gamma & -\frac{v}{c}\gamma & 0 \\ -\frac{v}{c}\gamma & \gamma & 0 \\ 0 & 0 & \overleftrightarrow{I}_\perp \end{bmatrix} \qquad L^\nu{}_{\mu'} = \begin{bmatrix} \gamma & \frac{v}{c}\gamma & 0 \\ \frac{v}{c}\gamma & \gamma & 0 \\ 0 & 0 & \overleftrightarrow{I}_\perp \end{bmatrix}$$

- $j^\mu = \begin{bmatrix} c\rho \\ j_\parallel \\ \vec{j}_\perp \end{bmatrix} \qquad \vec{v} = v\vec{e}_\parallel$

$$\Rightarrow \quad j^{\mu'} = \begin{bmatrix} c\rho' \\ j'_\parallel \\ \vec{j}'_\perp \end{bmatrix} = \underbrace{L^{\mu'}{}_\nu j^\nu}_{\text{maticové násobení}} = \begin{bmatrix} \gamma c(\rho - \frac{v}{c^2}j_\parallel) \\ \gamma(j_\parallel - v\rho) \\ \vec{j}_\perp \end{bmatrix}$$

$$\Rightarrow \quad \rho' = \gamma(\rho - \frac{v}{c^2}j_\parallel)$$
$$j'_\parallel = \gamma(j_\parallel - v\rho)$$
$$\vec{j}'_\perp = \vec{j}_\perp$$

- $A^{\mu'} = L^{\mu'}{}_\nu A^\nu \quad \Rightarrow \quad \phi' = \gamma(\phi - vA_\parallel)$

$$A'_\parallel = \gamma(A_\parallel - \frac{v}{c^2}\phi)$$
$$A'_\perp = A_\perp$$

Transformace $\vec{E}, \vec{B}$

$F'_{\mu'\nu'} = L^\kappa{}_{\mu'} L^\lambda{}_{\nu'} F_{\kappa\lambda}$

$F^{\mu'}{}_{\nu'} = L^{\mu'}{}_\kappa F^\kappa{}_\lambda L^\lambda{}_{\nu'}$ ← maticová interpretace

$$F^\mu{}_\nu = \begin{bmatrix} 0 & \frac{1}{c}\vec{E} \\ \frac{1}{c}\vec{E} & \overleftrightarrow{B} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{c}E_\parallel & \frac{1}{c}\vec{E}_\perp \\ \frac{1}{c}E_\parallel & 0 & \vec{B}_{\parallel\perp} \\ \frac{1}{c}\vec{E}_\perp & \vec{B}_{\perp\parallel} & \overleftrightarrow{B}_{\perp\perp} \end{bmatrix}$$

x $\quad$ y z

$$\overleftrightarrow{B} = \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \qquad \vec{B}_{\parallel\perp} = \vec{e}_\parallel\cdot\overleftrightarrow{B} = -\vec{e}_\parallel\times\vec{B}$$

$$= -\vec{e}_\parallel\times\vec{B}_\perp$$

$$\vec{B}_{\perp\parallel} = \vec{e}_\parallel\times\vec{B}_\perp$$

$$\overleftrightarrow{B}_{\perp\perp} = B_x\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = B_x\overleftrightarrow{\varepsilon}_\perp$$

$$F^{\mu'}{}_{\nu'} = \begin{bmatrix} \gamma & -\frac{v}{c}\gamma & 0 \\ -\frac{v}{c}\gamma & \gamma & 0 \\ 0 & 0 & \overleftrightarrow{I}_\perp \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{c}E_\parallel & \frac{1}{c}\vec{E}_\perp \\ \frac{1}{c}E_\parallel & 0 & \vec{B}_{\parallel\perp} \\ \frac{1}{c}\vec{E}_\perp & \vec{B}_{\perp\parallel} & \overleftrightarrow{B}_{\perp\perp} \end{bmatrix}\begin{bmatrix} \gamma & \frac{v}{c}\gamma & 0 \\ \frac{v}{c}\gamma & \gamma & 0 \\ 0 & 0 & \overleftrightarrow{I}_\perp \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{v}{c}\gamma E_\parallel & \gamma\frac{1}{c}E_\parallel & \gamma\frac{1}{c}(\vec{E}_\perp - v\vec{B}_{\parallel\perp}) \\ \gamma\frac{1}{c}E_\parallel & -\gamma\frac{v}{c^2}E_\parallel & \gamma(\vec{B}_{\parallel\perp} - \frac{v}{c^2}\vec{E}_\perp) \\ \frac{1}{c}\vec{E}_\perp & \vec{B}_{\perp\parallel} & \overleftrightarrow{B}_{\perp\perp} \end{bmatrix}\begin{bmatrix} & & \\ & \cdot & \\ & & \cdot \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{1}{c}E_\parallel & \gamma\frac{1}{c}(\vec{E}_\perp - v\vec{B}_{\parallel\perp}) \\ \frac{1}{c}E_\parallel & 0 & \gamma(\vec{B}_{\parallel\perp} - \frac{v}{c^2}\vec{E}_\perp) \\ \frac{1}{c}\gamma(\vec{E}_\perp + v\vec{B}_{\perp\parallel}) & \gamma(\vec{B}_{\perp\parallel} + \frac{v}{c^2}\vec{E}_\perp) & \overleftrightarrow{B}_\perp \end{bmatrix}$$

$\Rightarrow \quad E'_\parallel = E_\parallel \qquad \vec{E}'_\perp = \gamma(\vec{E}_\perp + \vec{v}\times\vec{B}_\perp)$

$\overleftrightarrow{B}'_\perp = \overleftrightarrow{B}_\perp \qquad \vec{B}'_{\perp\parallel} = \gamma(\vec{B}_{\perp\parallel} + \frac{v}{c^2}\vec{E}_\perp) \;\Big|\; -\vec{e}_\parallel\times$ zprava

$B'_\parallel = B_\parallel \qquad \downarrow$

$$\vec{B}'_\perp = \gamma(\vec{B}_\perp - \frac{1}{c^2}\vec{v}\times\vec{E}_\perp)$$

=> míchají se $\vec{E}$ a $\vec{B}$ -> jsou to součásti stejné věci pozorované v různých soustavách

2 invarianty:

$\mathcal{L} \sim E^2 - c^2B^2$

$\tilde{\mathcal{L}} \sim \vec{E}\cdot\vec{B}$

1) $E = cB \quad \Rightarrow \quad E' = cB'$

2) $\vec{E}\cdot\vec{B} = 0 \quad \Rightarrow \quad \vec{E}'\cdot\vec{B} = 0$

3) $\vec{B} = 0 \qquad \tilde{\mathcal{L}} = 0 \qquad \mathcal{L} > 0$

$\vec{E}'$ musí být ⊥ na $\vec{B}'$ a $\mathcal{L} > 0$ ⇓ nemůže být $\vec{E} = \vec{0}$

$\Rightarrow \quad \vec{E}' = E_\parallel\vec{e}_\parallel + \gamma\vec{E}_\perp$

$\vec{B}' = -\gamma\frac{1}{c^2}\vec{v}\times\vec{E}_\perp$

$= -\frac{1}{c^2}\vec{v}\times\vec{E}'$

4) $\vec{E} = 0 \qquad \tilde{\mathcal{L}} = 0 \qquad \mathcal{L} < 0$

pole na sebe budou již kolmá

$\vec{B}' = B_\parallel\vec{e}_\parallel + \gamma\vec{B}_\perp$

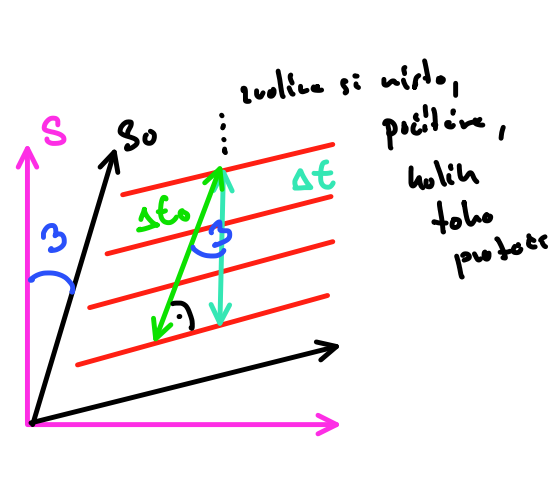
$\vec{E}' = \vec{v}\times\vec{B}'$

Přímý vodič

$\lambda = \frac{\text{náboj}}{\text{délka}} \quad \Rightarrow \quad \lambda = \gamma\lambda_0$

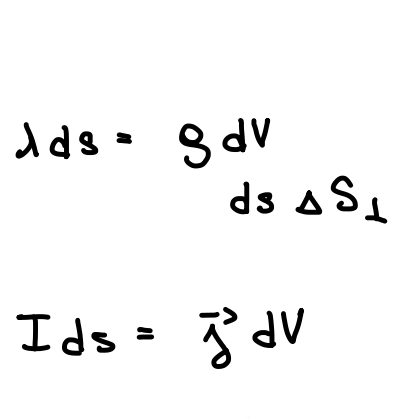
$I = \frac{\text{náboj}}{\text{čas}} \quad \Rightarrow \quad I = \gamma I_0$ $\qquad$ rychlá úvaha

pohybuje se na druhou stranu



z geometrie trojúhelníka ↓

$\frac{\Delta s_0}{\Delta s} = \cosh\vartheta = \gamma$

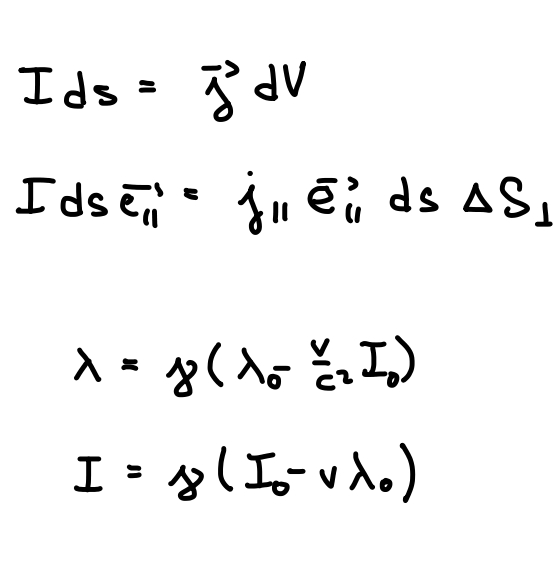


nejdříve vidíme Δs v tomto obrázku

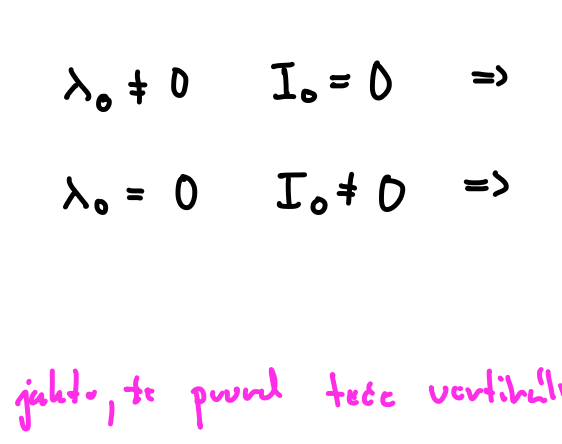
v tomto místě a počítám, kolik nábojů proteče

tedy vidím čas

tedy jsem proud, musím čítat náboj



$\frac{\Delta t_0}{\Delta t} = \cosh\vartheta = \gamma$



⇓ jako měřítko vzdálenost / čas, náboj je invariantní

$\lambda ds = \rho dV$

$\qquad ds\,\Delta S_\perp$

$I ds = \vec{j}\,dV \qquad \begin{bmatrix} c\lambda \\ I \end{bmatrix} \leftrightarrow \begin{bmatrix} c\rho \\ j_\parallel \end{bmatrix}$

$I ds\,\vec{e}_\parallel = j_\parallel\vec{e}_\parallel\,ds\,\Delta S_\perp$

$\lambda = \gamma(\lambda_0 - \frac{v}{c^2}I_0)$

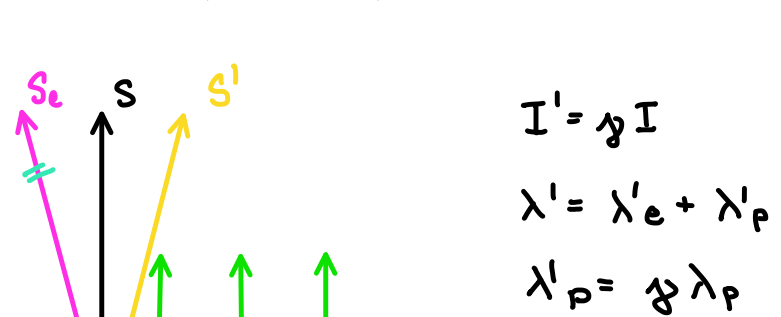
$I = \gamma(I_0 - v\lambda_0)$

$\lambda_0 \neq 0 \quad I_0 = 0 \quad \Rightarrow \quad \lambda = \gamma\lambda_0 \qquad I = -\gamma v\lambda_0 = -v\lambda$

$\lambda_0 = 0 \quad I_0 \neq 0 \quad \Rightarrow \quad \lambda = -\gamma\frac{v}{c^2}I_0 \quad I = \gamma I_0$

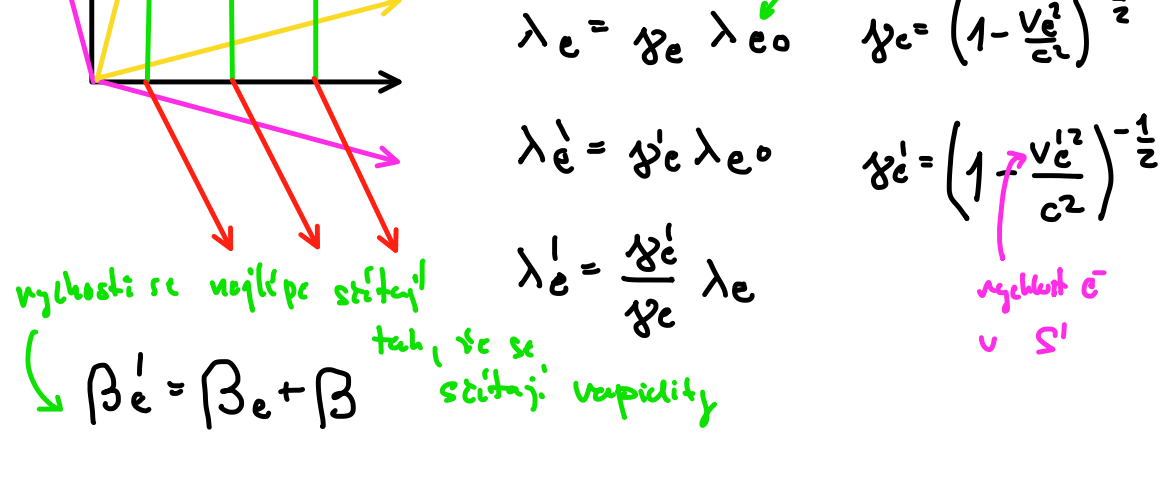
$= -\frac{v}{c^2}I$

jakto, že proud teče vertikálně? - Nemusí být vždy jen kauzální, co znamená neutrální drát s proudem?

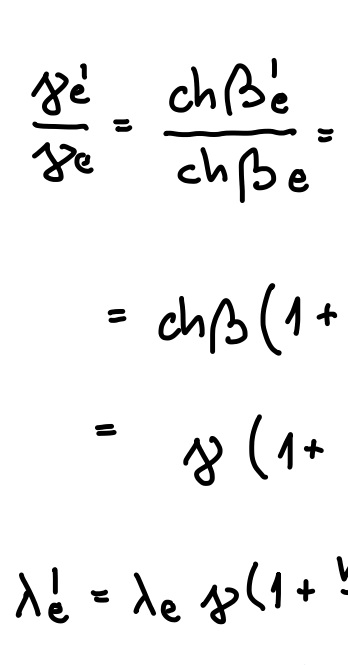


$\lambda = \lambda_e + \lambda_p = 0$

$I = I_e + I_p = I_e = -v_e\lambda_e$



j_p^μ ← vlastní λ_p - není jiná složka
I
nemusí spolu v čase, jen proud má opačný směr, protože mají záporný náboj
j_e^μ
pohybují se podsvětelně



$I' = \gamma I$

$\lambda' = \lambda'_e + \lambda'_p$

$\lambda'_p = \gamma\lambda_p$ $\qquad$ vlastní soustava e^- => dopočítám, kolik tam bude vidět

$\lambda_e = \gamma_e\lambda_{e0} \qquad \gamma_e = \left(1 - \frac{v_e^2}{c^2}\right)^{-\frac{1}{2}}$

$\lambda'_e = \gamma'_e\lambda_{e0} \qquad \gamma'_e = \left(1 - \frac{v_e'^2}{c^2}\right)^{-\frac{1}{2}}$ ← rychlost e^- v S'

$\lambda'_e = \frac{\gamma'_e}{\gamma_e}\lambda_e$

rychlosti se nejlépe sčítají tak, že se sčítají rapidity

$\beta'_e = \beta_e + \beta$

$$\frac{\gamma'_e}{\gamma_e} = \frac{\mathrm{ch}\beta'_e}{\mathrm{ch}\beta_e} = \frac{\mathrm{ch}\beta_e\mathrm{ch}\beta + \mathrm{sh}\beta\,\mathrm{sh}\beta_e}{\mathrm{ch}\beta_e}$$

$$= \mathrm{ch}\beta(1 + \mathrm{th}\beta_e\mathrm{th}\beta)$$

$$= \gamma\left(1 + \frac{v_e v}{c^2}\right)$$

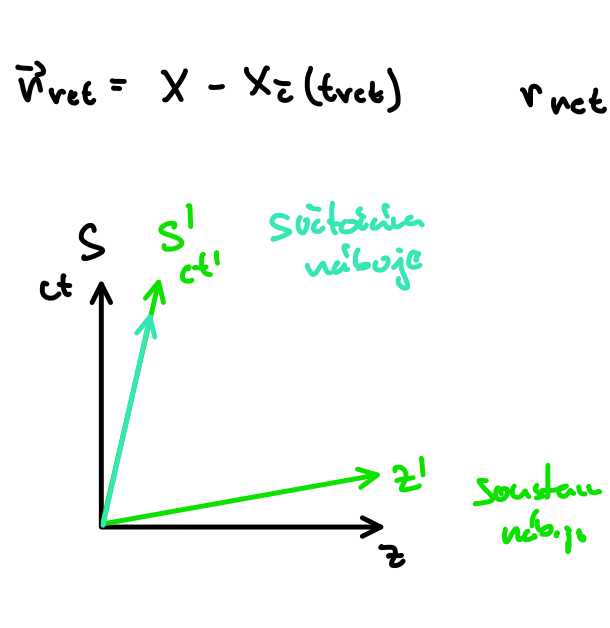
$$\lambda'_e = \lambda_e\gamma\left(1 + \frac{v_e v}{c^2}\right)$$

$$\lambda' = \lambda_e\gamma\left(\cancel{1} + \frac{v_e v}{c^2}\right) + \cancel{\lambda_p\gamma} = \overbrace{\gamma\underbrace{\lambda_e v_e}_{-I}}^{-I'}\frac{v}{c^2}$$

$$= -\frac{v}{c^2}I'$$

lokální hustota elektronů a protonů není stejná!

Rovnoměrně pohybující se náboj

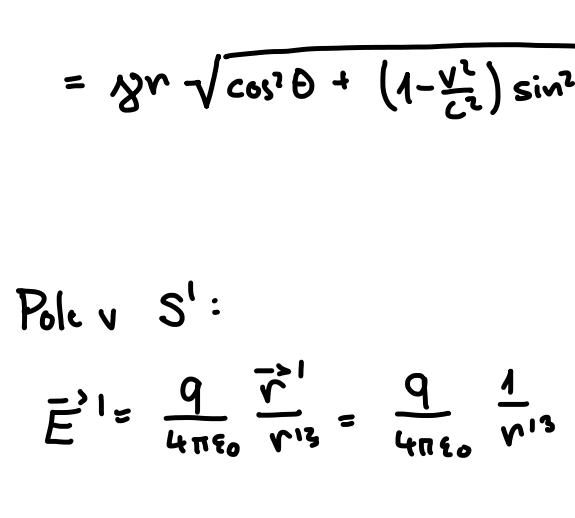


$X_z(t_{ret}) \qquad z_z(t) = vt \qquad x_z(t) = y_z(t) = 0$

$\vec{r}(t) = X - X_z(t) = (z - vt)\vec{e}_\parallel + R\vec{e}_R$

$= r(t)\,(\cos\theta\,\vec{e}_\parallel + \sin\theta\,\vec{e}_R) \qquad \mathrm{tg}\,\theta = \frac{R}{z - vt}$

$\vec{r}_{ret} = X - X_z(t_{ret}) \qquad r_{ret} = c(t - t_{ret})$



$t' = \gamma(t - \frac{v}{c^2}z)$

$z' = \gamma(z - vt)$

$R' = R$

$\vec{r}' = X' - X'_z = X' - P'$

$$r' = \sqrt{R'^2 + z'^2} = \sqrt{\gamma^2(z - vt)^2 + R^2}$$

$$= \gamma r\sqrt{\cos^2\theta + \left(1 - \frac{v^2}{c^2}\right)\sin^2\theta} = \gamma r\sqrt{1 - \frac{v^2}{c^2}\sin^2\theta}$$

Pole v S':

$$\vec{E}' = \frac{q}{4\pi\varepsilon_0}\frac{\vec{r}'}{r'^3} = \frac{q}{4\pi\varepsilon_0}\frac{1}{r'^3}(z'\vec{e}'_\parallel + R'\vec{e}'_R)$$

$$E_\parallel = \frac{q}{4\pi\varepsilon_0}\frac{z'}{r'^3} \qquad \vec{E}_\perp = \frac{q}{4\pi\varepsilon_0}\frac{R'}{r'^3}\vec{e}_R$$

$\vec{B}' \equiv \vec{0}$

Pole v S - Lorentzova transformace s rychlostí $-v$

$$E_\parallel = E'_\parallel = \frac{q}{4\pi\varepsilon_0}\frac{\gamma(z - vt)}{r'^3}$$

$$\vec{E}_\perp = \gamma\vec{E}'_\perp + \text{příspěvek od } \vec{B} \Rightarrow \text{je nulový} = \frac{q}{4\pi\varepsilon_0}\frac{\gamma R}{r'^3}\vec{e}_R$$

$$\Rightarrow \quad \vec{E} = \frac{q}{4\pi\varepsilon_0}\frac{\gamma}{r'^3}\underbrace{\left((z - vt)\vec{e}_\parallel + R\vec{e}_R\right)}$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{\gamma}{r'^3}\vec{r}(t)$$

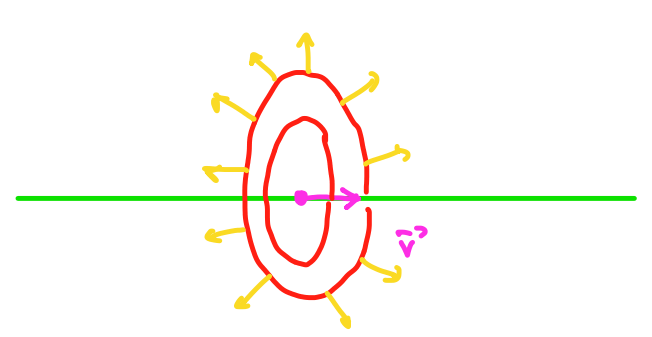
$$= \frac{q}{4\pi\varepsilon_0}\overbrace{\frac{1}{\gamma^2}}^{= 1 - \frac{v^2}{c^2}}\frac{1}{\left(1 - \frac{v^2}{c^2}\sin^2\theta\right)^{\frac{3}{2}}}\frac{\vec{e}(t)}{r^2(t)}$$

příroda se snaží udělat aproximaci rovnoměrným přímočarým pohybem ← Směr je dán vektorem v čase t

$B_\parallel = B'_\parallel = 0$

$$\vec{B}_\perp = \gamma\frac{1}{c^2}\vec{v}\times\vec{E}'_\perp = \frac{1}{c^2}\vec{v}\times\vec{E}_\perp = \frac{1}{c^2}\vec{v}\times\vec{E}$$

$$\vec{B} = \frac{1}{c^2}\vec{v}\times\vec{E} = \frac{qv}{4\pi\varepsilon_0 c^2}\frac{\left(1 - \frac{v^2}{c^2}\right)\sin\theta}{\left(1 - \frac{v^2}{c^2}\sin^2\theta\right)^{\frac{3}{2}}}\frac{\vec{e}_\varphi}{r^2(t)}$$

← z vektorového součinu